

```
-c -o user/sixfive.o user/sixfive.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_sixfive user/_sixfive.o user/_ulib.o user/_usys.o user/_printf.o user/_unalloc.o
riscv64-linux-gnu-objdump -S user/_sixfive > user/sixfive.asm
riscv64-linux-gnu-objdump -t user/_sixfive | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/sixfive.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026=. -DSOL_UTIL -DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mmodel=medan
y -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-st
rchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie
-c -o user/mendump.o user/mendump.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_mendump user/_mendump.o user/_ulib.o user/_usys.o user/_printf.o user/_unalloc.o
riscv64-linux-gnu-objdump -S user/_mendump > user/mendump.asm
riscv64-linux-gnu-objdump -t user/_mendump | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/mendump.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026=. -DSOL_UTIL -DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mmodel=medan
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rchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie
-c -o user/find.o user/find.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_find user/_find.o user/_ulib.o user/_usys.o user/_printf.o user/_unalloc.o
riscv64-linux-gnu-objdump -S user/_find > user/find.asm
riscv64-linux-gnu-objdump -t user/_find | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/find.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026=. -DSOL_UTIL -DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mmodel=medan
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rchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie
-c -o user/uptime.o user/uptime.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_uptime user/_uptime.o user/_ulib.o user/_usys.o user/_printf.o user/_unalloc.o
riscv64-linux-gnu-objdump -S user/_uptime > user/uptime.asm
riscv64-linux-gnu-objdump -t user/_uptime | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/uptime.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026=. -DSOL_UTIL -DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mmodel=medan
y -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-st
rchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie
-c -o user/freemem.o user/freemem.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_freemem user/_freemem.o user/_ulib.o user/_usys.o user/_printf.o user/_unalloc.o
riscv64-linux-gnu-objdump -S user/_freemem > user/freemem.asm
riscv64-linux-gnu-objdump -t user/_freemem | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/freemem.sym
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertestfs user/_g
find user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sync user/_sleep user/_sixfive user/_mendump user/_find user/_uptime user/_freemem user/_mendump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2008
ballocc: first 1274 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -n 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bu
s=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ freemem
Free memory: 133234688 bytes
$
```